Term 3 Practice Activity: C1 Density Functions

Contents

Evaluated criteria

- C1. Describes probabilities in continuous variables as a density function.

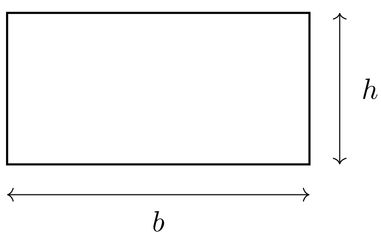
1 Areas of Basic Shapes

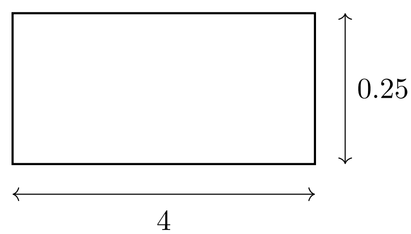
Problem. Find the area of each figure using the measurements shown. In each case, also answer the corresponding general question with variable lengths.

1. Rectangle

Specific question. Find the area of the following rectangle.

General question. If a rectangle has base b and height h, what is its area?





Solution. For a rectangle with base b and height h,

$$A = bh.$$

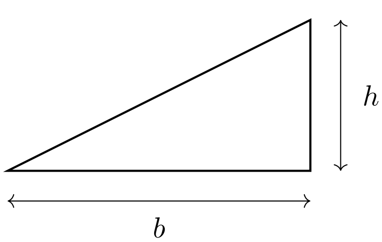
For the given rectangle,

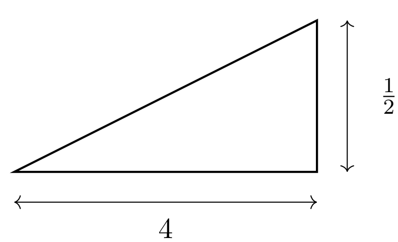
$$A = bh = 4(0.25) = 1.$$

2. Triangle

Specific question. Find the area of the following triangle.

General question. If a triangle has base b and height h, what is its area?





Solution. For a triangle with base b and height h,

$$A = \frac{1}{2}bh.$$

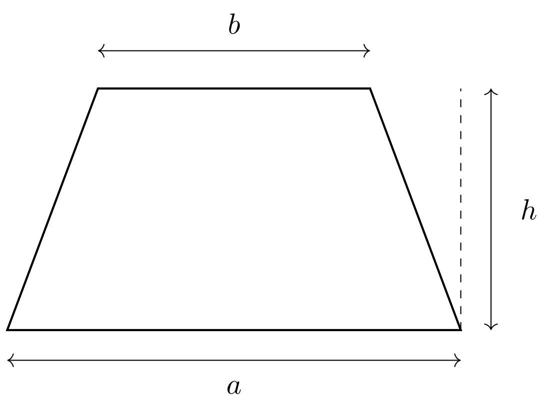
For the given triangle,

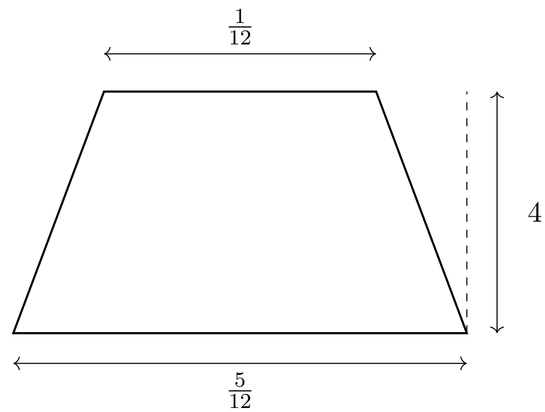
$$A = \frac{1}{2}bh = \frac{1}{2}(4)\left(\frac{1}{2}\right) = 1.$$

3. Trapeze

Specific question. Find the area of the following trapeze.

General question. If a trapeze has parallel sides of lengths a and b, and height h, what is its area?





Solution. For a trapeze with parallel sides a and b, and height h,

$$A = \frac{a+b}{2}h.$$

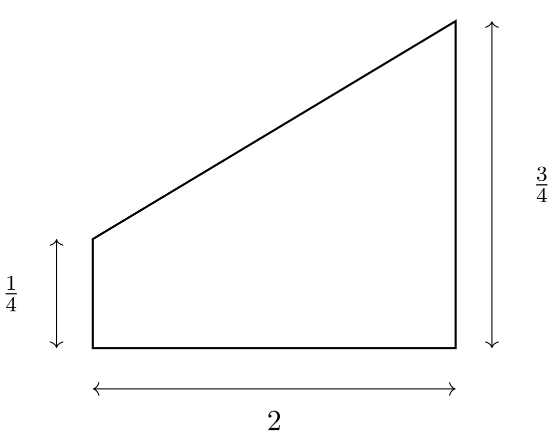
For the given trapeze,

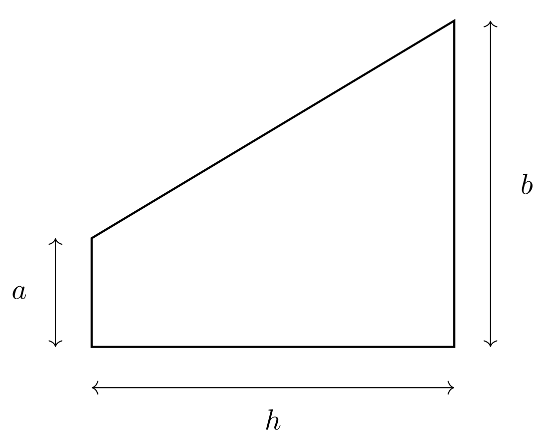
$$A = \frac{a+b}{2}h = \frac{\frac{5}{12}+\frac{1}{12}}{2}\cdot 4 = \frac{\frac{6}{12}}{2}\cdot 4 = \frac{\frac{1}{2}}{2}\cdot 4 = \frac{1}{4}\cdot 4 = 1.$$

4. Horizontal Trapeze

Specific question. Find the area of the following trapeze.

General question. If a horizontal trapeze has parallel sides of lengths a and b, and horizontal distance h between them, what is its area?





Solution. For the given trapeze,

$$A = \frac{a+b}{2}h = \frac{\frac{1}{4}+\frac{3}{4}}{2} \cdot 2 = \frac{1}{2} \cdot 2 = 1.$$

For a horizontal trapeze with parallel sides a and b, and horizontal distance h between them,

$$A = \frac{a+b}{2}h.$$

2 Checking Two Candidate Density Functions I

Problem. A continuous random variable is modeled on the interval shown in each case.

Distribution A

$$f(x) = 0.25 \quad \text{for } 0 \le x \le 4$$

Distribution B

$$f(x) = 0.40 \quad \text{for } 0 \le x \le 4$$

Questions/Tasks.

- **C1 Calculations:** Decide which distribution is a valid PDF and which one is not.
- **C1 Interpretation:** Justify your answer using the total area under each graph.
- **Extra analysis:** For the valid distribution, find $P(1 \le X \le 3)$.

C1

C1 calculations For a valid PDF, $f(x) \ge 0$ on its support and total area is 1.

- Distribution A: support $[0,4]$, constant height 0.25.

  $$\text{Area} = 4(0.25) = 1,$$

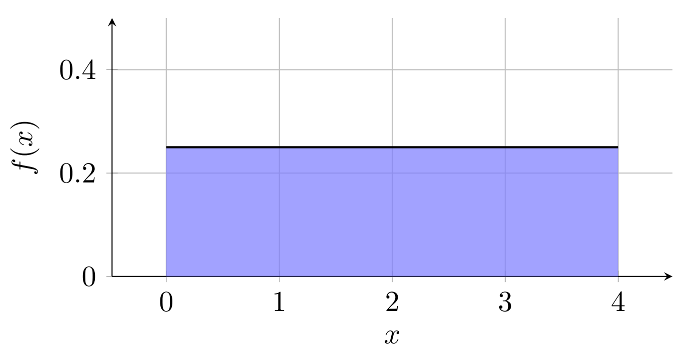
  so A is valid.

- Distribution B: support $[0,4]$, constant height 0.40.

  $$\text{Area} = 4(0.40) = 1.6 \ne 1,$$

  so B is not valid.

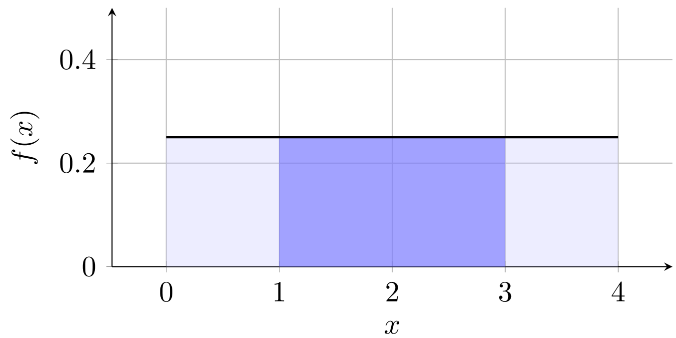
C1 interpretation



The valid PDF is uniform on $[0, 4]$: it is constant, nonnegative, and gives equal probability to equal-width intervals.

Extra analysis. For the valid distribution A,

$$P(1 \le X \le 3) = (3 - 1)(0.25) = 0.5.$$



Final answer: A is valid, B is not valid, and $P(1 \le X \le 3) = 0.5$.

3 Checking Two Candidate Density Functions II

Problem. A continuous random variable is modeled on the interval shown in each case.

Distribution A

$$f(x) = \frac{x}{8} \quad \text{for } 0 \le x \le 4$$

Distribution B

$$f(x) = \frac{x}{6} \quad \text{for } 0 \le x \le 4$$

Questions/Tasks.

- **C1 Calculations:** Decide which distribution is a valid PDF and which one is not.
- **C1 Interpretation:** Justify your answer using geometric reasoning about the area under each line.
- **Extra analysis:** For the valid distribution, find $P(2 \le X \le 4)$.

C1

C1 calculations Check nonnegativity and total area on support $[0, 4]$.

- Distribution A: $f(x) = \frac{x}{8}$ forms a triangle with base 4 and height $\frac{4}{8} = \frac{1}{2}$.

$$\text{Area} = \frac{1}{2}(4)\left(\frac{1}{2}\right) = 1,$$

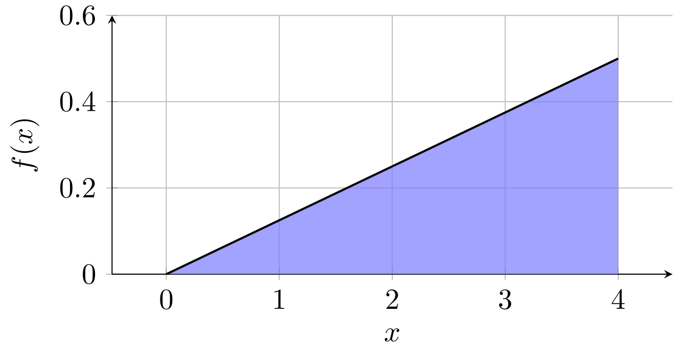
  so A is valid.

- Distribution B: $f(x) = \frac{x}{6}$ has base 4 and height $\frac{4}{6} = \frac{2}{3}$.

$$\text{Area} = \frac{1}{2}(4)\left(\frac{2}{3}\right) = \frac{4}{3} \neq 1,$$

  so B is not valid.

C1 interpretation



The valid PDF increases linearly on $[0, 4]$, so larger x values have higher density than smaller ones.

Extra analysis. For A,

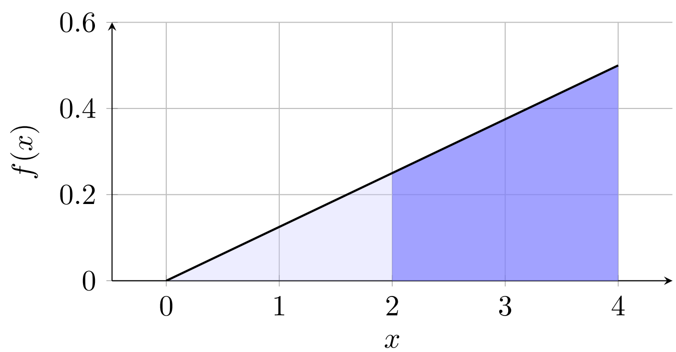
$$P(2 \leq X \leq 4) = 1 - P(0 \leq X \leq 2).$$

On $[0, 2]$, area is a triangle with base 2 and height $\frac{2}{8} = \frac{1}{4}$:

$$P(0 \leq X \leq 2) = \frac{1}{2}(2)\left(\frac{1}{4}\right) = \frac{1}{4}.$$

Hence

$$P(2 \leq X \leq 4) = 1 - \frac{1}{4} = \frac{3}{4}.$$



Final answer: A is valid, B is not valid, and $P(2 \leq X \leq 4) = \frac{3}{4}$.

4 Checking Two Candidate Density Functions III

Problem. A continuous random variable is modeled on the interval shown in each case.

Distribution A

$$f(x) = \frac{6 - x}{12} \quad \text{for } 1 \leq x \leq 5$$

Distribution B

$$f(x) = \frac{6 - x}{10} \quad \text{for } 1 \leq x \leq 5$$

Questions/Tasks.

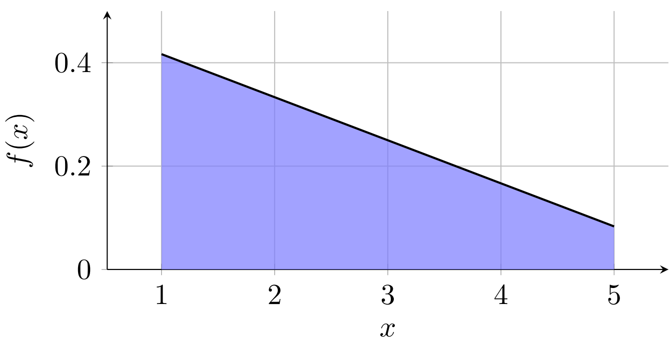
- **C1 Calculations:** Decide which distribution is a valid PDF and which one is not.
- **C1 Interpretation:** Justify your answer using geometric reasoning about the area under each line.
- **Extra analysis:** For the valid distribution, find $P(1 \le X \le 3)$.

C1

C1 calculations On support $[1, 5]$, both densities are nonnegative. Check total area:

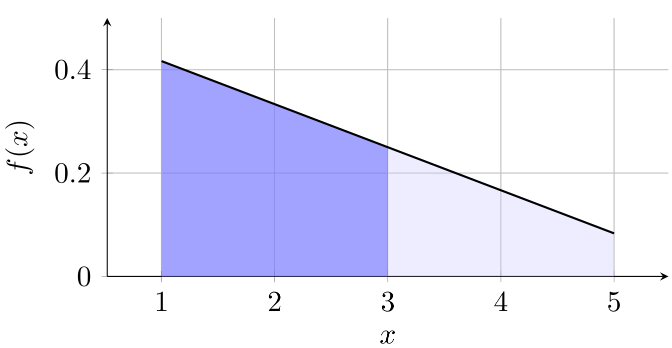
- Distribution A: $f(x) = \frac{6-x}{12}$ gives a trapezium with base 4, left height $f(1) = \frac{5}{12}$, and right height $f(5) = \frac{1}{12}$.
$$\text{Area} = \frac{\frac{5}{12} + \frac{1}{12}}{2} \cdot 4 = 1,$$
so A is valid.
- Distribution B: $f(x) = \frac{6-x}{10}$ gives base 4, left height $f(1) = \frac{5}{10} = \frac{1}{2}$, and right height $f(5) = \frac{1}{10}$.
$$\text{Area} = \frac{\frac{1}{2} + \frac{1}{10}}{2} \cdot 4 = \frac{6}{5} \neq 1,$$
so B is not valid.

C1 interpretation



The valid PDF is decreasing on $[1, 5]$, so values near 1 are more likely than values near 5, but values near 5 still have positive density.

Extra analysis. For valid distribution A,
$$P(1 \le X \le 3) = \text{area of trapezium on } [1, 3].$$
With heights $f(1) = \frac{5}{12}$ and $f(3) = \frac{1}{4}$, width 2:
$$P(1 \le X \le 3) = \frac{\frac{5}{12} + \frac{1}{4}}{2} \cdot 2 = \frac{2}{3}.$$



Final answer: A is valid, B is not valid, and $P(1 \le X \le 3) = \frac{2}{3}$.

5 Water Storage Level (Uniform)

Problem. Let X be the water storage level (in m) with support $[0, 4]$ and density

$$f(x) = k \quad \text{for } 0 \le x \le 4.$$

Questions/Tasks.

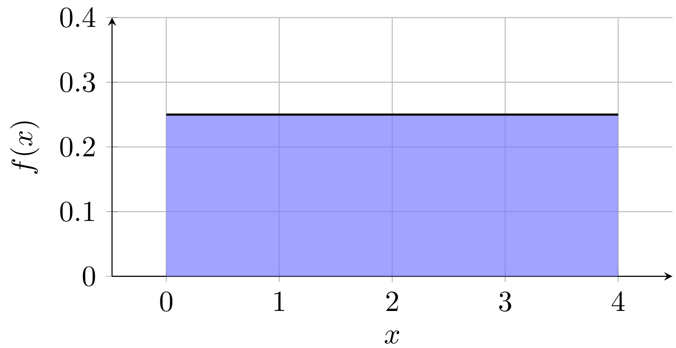
- **C1 Calculations:** Find k so $f(x)$ is a valid PDF.
- **C1 Interpretation:** Explain what this model says about equal-length intervals.
- **Extra analysis:** Find $P(1 \le X \le 3)$.

C1

C1 calculations Since $f(x) = k$ on $[0, 4]$, total area is rectangle area:

$$4k = 1 \implies k = \frac{1}{4}.$$

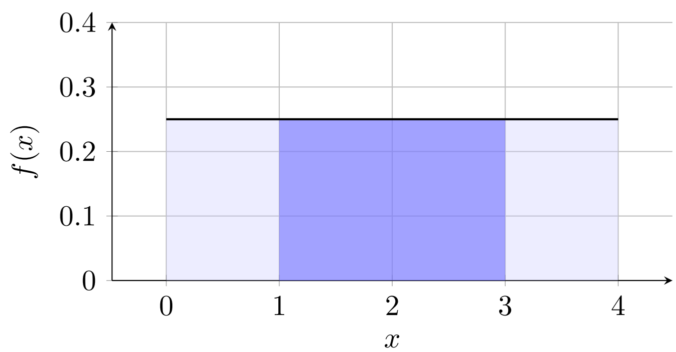
C1 interpretation



The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. For the requested probability,

$$P(1 \le X \le 3) = \text{width} \times \text{height} = (3 - 1)\left(\frac{1}{4}\right) = \frac{1}{2}.$$



Final answer: $k = \frac{1}{4}$ and $P(1 \le X \le 3) = \frac{1}{2}$.

6 Startup Delay (Linear, Endpoint at 0)

Problem. Let T be a startup delay (in s) with support $[0, 6]$ and density

$$f(t) = kt \quad \text{for } 0 \le t \le 6.$$

Questions/Tasks.

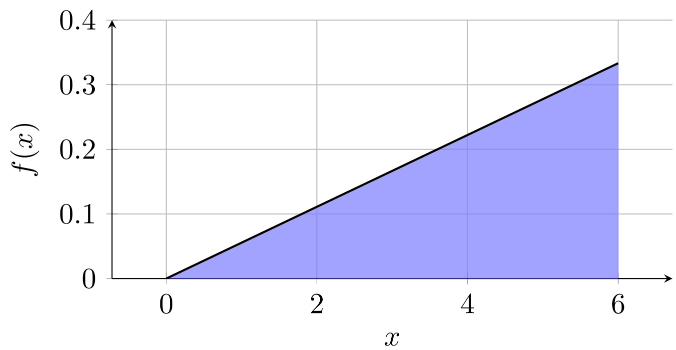
- **C1 Calculations:** Find k so $f(t)$ is a valid PDF.
- **C1 Interpretation:** Describe how probability changes as t increases.
- **Extra analysis:** Find $P(T \ge 4)$.

C1

C1 calculations For $f(t) = kt$ on $[0, 6]$, area is a triangle with base 6 and height $6k$:

$$\frac{1}{2}(6)(6k) = 18k = 1 \implies k = \frac{1}{18}.$$

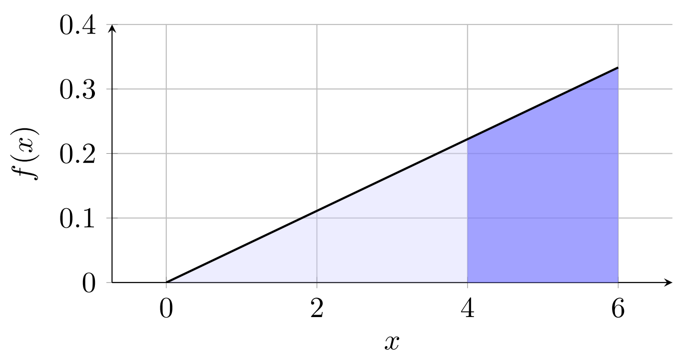
C1 interpretation



The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. For $P(T \ge 4)$, use area from 4 to 6 (a trapezium): heights are $4k$ and $6k$, width 2.

$$P(T \ge 4) = \frac{4k + 6k}{2} \cdot 2 = 10k = \frac{10}{18} = \frac{5}{9}.$$



Final answer: $k = \frac{1}{18}$ and $P(T \geq 4) = \frac{5}{9}$.

7 Packing Score (Linear, Both Endpoints Nonzero)

Problem. Let Q be a packing score with support $[2, 8]$ and density

$$f(q) = k(q-1) \quad \text{for } 2 \leq q \leq 8.$$

Questions/Tasks.

- **C1 Calculations:** Find k so $f(q)$ is a valid PDF.
- **C1 Interpretation:** Compare density near $q = 3$ and $q = 7$.
- **Extra analysis:** Find $P(4 \leq Q \leq 6)$.

C1

C1 calculations On $[2, 8]$, $f(q) = k(q-1)$ is linear and positive. Total area is a trapezium with width 6, heights

$$f(2) = k, \quad f(8) = 7k.$$

So

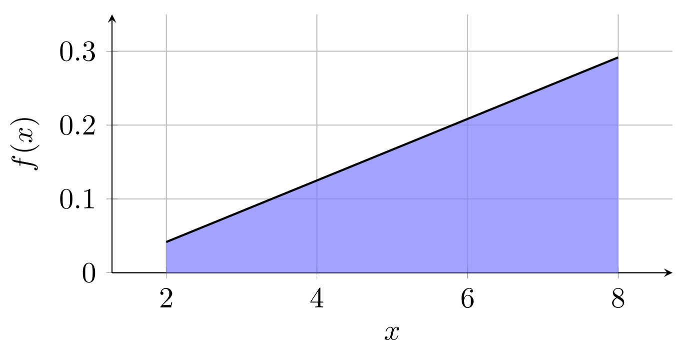
$$\frac{k + 7k}{2} \cdot 6 = 24k = 1 \implies k = \frac{1}{24}.$$

Compare densities:

$$f(3) = 2k, \quad f(7) = 6k,$$

so density near $q = 7$ is three times density near $q = 3$.

C1 interpretation



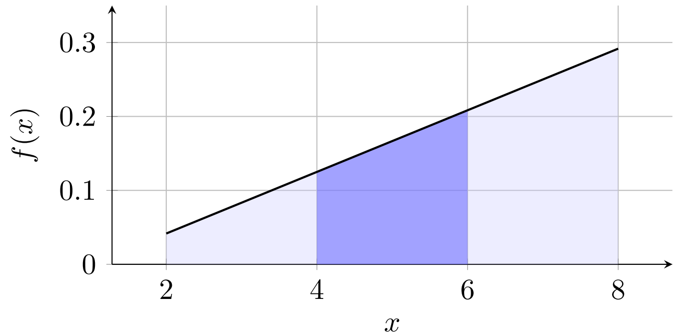
The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. For

$$P(4 \leq Q \leq 6) = \text{trapezium area on } [4, 6],$$

with heights $f(4) = 3k$, $f(6) = 5k$, width 2:

$$P(4 \leq Q \leq 6) = \frac{3k + 5k}{2} \cdot 2 = 8k = \frac{1}{3}.$$



Final answer: $k = \frac{1}{24}$ and $P(4 \le Q \le 6) = \frac{1}{3}$.

8 Classroom Temperature Band (Uniform)

Problem. Let Y be the classroom temperature offset (in °C) with support $[-2, 3]$ and density

$$f(y) = k \quad \text{for } -2 \le y \le 3.$$

Questions/Tasks.

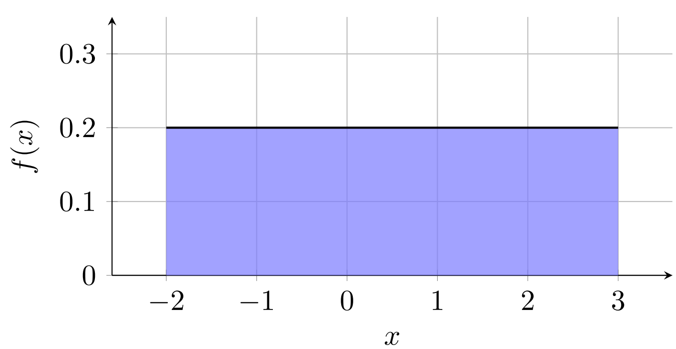
- **C1 Calculations:** Find k so $f(y)$ is a valid PDF.
- **C1 Interpretation:** Explain why interval length controls probability here.
- **Extra analysis:** Find $P(-1 \le Y \le 2)$.

C1

C1 calculations Uniform on $[-2, 3]$ gives length 5, so

$$5k = 1 \implies k = \frac{1}{5}.$$

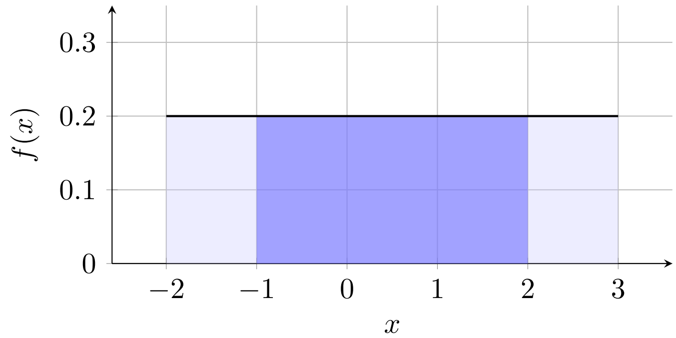
C1 interpretation



The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. Since density is constant, probability depends only on interval length:

$$P(-1 \le Y \le 2) = (2 - (-1)) \cdot \frac{1}{5} = \frac{3}{5}.$$



Final answer: $k = \frac{1}{5}$ and $P(-1 \le Y \le 2) = \frac{3}{5}$.

9 Assembly Time (Linear, Endpoint at 0)

Problem. Let A be assembly time (in min) with support $[0, 5]$ and density

$$f(a) = k(5 - a) \quad \text{for } 0 \le a \le 5.$$

Questions/Tasks.

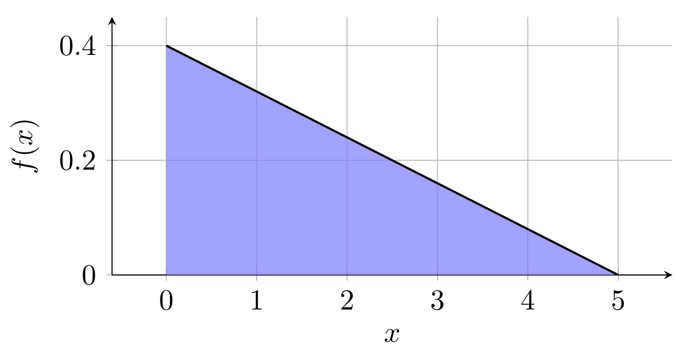
- **C1 Calculations:** Find k so $f(a)$ is a valid PDF.
- **C1 Interpretation:** Explain what the decreasing line says about short vs long times.
- **Extra analysis:** Find $P(1 \le A \le 4)$.

C1

C1 calculations $f(a) = k(5 - a)$ on $[0, 5]$ gives a decreasing line from height $5k$ to 0. Total area (triangle):

$$\frac{1}{2}(5)(5k) = \frac{25k}{2} = 1 \implies k = \frac{2}{25}.$$

C1 interpretation



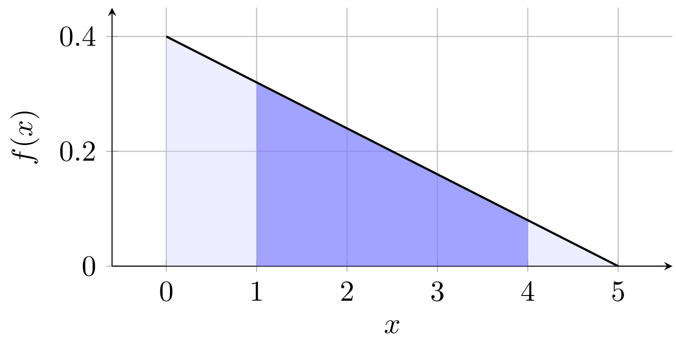
The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. For $1 \le A \le 4$, use trapezium area with heights

$$f(1) = 4k, \quad f(4) = k,$$

and width 3:

$$P(1 \le A \le 4) = \frac{4k + k}{2} \cdot 3 = \frac{15k}{2} = \frac{3}{5}.$$



Final answer: $k = \frac{2}{25}$ and $P(1 \le A \le 4) = \frac{3}{5}$.

10 Quality Index (Linear, Both Endpoints Nonzero)

Problem. Let R be a quality index with support $[1, 7]$ and density

$$f(r) = k(r + 1) \quad \text{for } 1 \le r \le 7.$$

Questions/Tasks.

- **C1 Calculations:** Find k so $f(r)$ is a valid PDF.
- **C1 Interpretation:** Explain why right-side intervals have more probability than equal-width left-side intervals.
- **Extra analysis:** Find $P(R \le 4)$.

C1

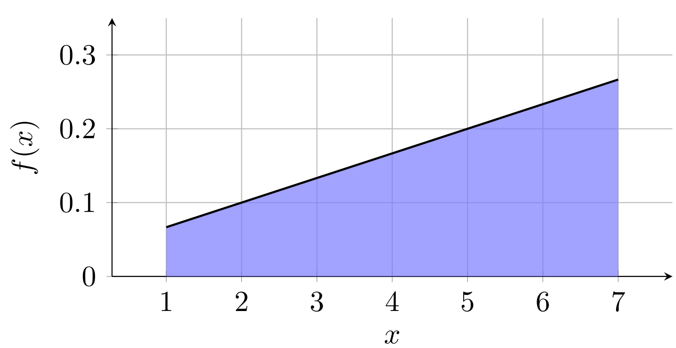
C1 calculations On $[1, 7]$, $f(r) = k(r + 1)$ has

$$f(1) = 2k, \quad f(7) = 8k.$$

Total area (trapezium, width 6):

$$\frac{2k + 8k}{2} \cdot 6 = 30k = 1 \implies k = \frac{1}{30}.$$

C1 interpretation



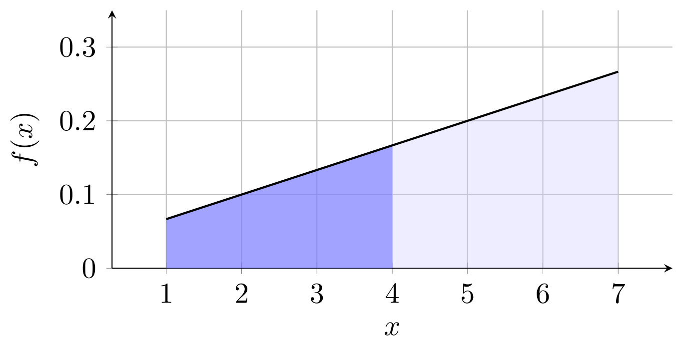
The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. For

$$P(R \le 4) = P(1 \le R \le 4) = \frac{f(1) + f(4)}{2} \cdot (4 - 1),$$

with $f(4) = 5k$:

$$P(R \le 4) = \frac{2k + 5k}{2} \cdot 3 = \frac{21k}{2} = \frac{7}{20}.$$



Final answer: $k = \frac{1}{30}$ and $P(R \le 4) = \frac{7}{20}$.

11 Rainfall Amount (Uniform)

Problem. Let M be rainfall amount (in mm) with support $[3, b]$ and density

$$f(m) = \frac{1}{6} \quad \text{for } 3 \le m \le b.$$

Questions/Tasks.

- **C1 Calculations:** Find b so $f(m)$ is a valid PDF.
- **C1 Interpretation:** State two different intervals with equal probability and justify.
- **Extra analysis:** Find $P(4.5 \le M \le 7.5)$.

C1

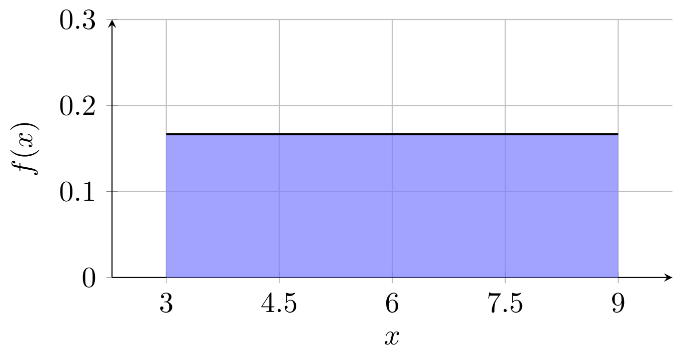
C1 calculations The height is fixed at $\frac{1}{6}$, so the total area must be

$$(b - 3) \cdot \frac{1}{6} = 1.$$

Hence

$$b - 3 = 6 \implies b = 9.$$

C1 interpretation

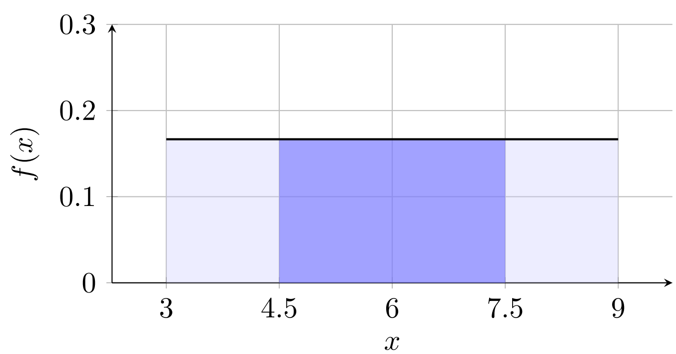


The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. Two equal-length intervals have equal probability, for example:

$$P(3 \le M \le 5) = P(7 \le M \le 9) = 2 \cdot \frac{1}{6} = \frac{1}{3}.$$

$$P(4.5 \le M \le 7.5) = (7.5 - 4.5) \cdot \frac{1}{6} = \frac{3}{6} = \frac{1}{2}.$$



Final answer: $b = 9$, equal-length intervals such as $[3, 5]$ and $[7, 9]$ have equal probability, and $P(4.5 \le M \le 7.5) = \frac{1}{2}$.

12 Battery Discharge (Linear, Endpoint at 0)

Problem. Let B be battery discharge measure with support $[0, c]$ and density

$$f(b) = \frac{1}{16} + \frac{b}{8c} \quad \text{for } 0 \le b \le c.$$

Questions/Tasks.

- **C1 Calculations:** Find c so $f(b)$ is a valid PDF.
- **C1 Interpretation:** Compare likelihood of values in $[0, 2]$ and $[6, 8]$.
- **Extra analysis:** Find $P(2 \le B \le 6)$.

C1

C1 calculations The endpoint values are fixed:

$$f(0) = \frac{1}{16}, \quad f(c) = \frac{1}{16} + \frac{c}{8c} = \frac{3}{16}.$$

Total area must be 1. Using the trapezium formula,

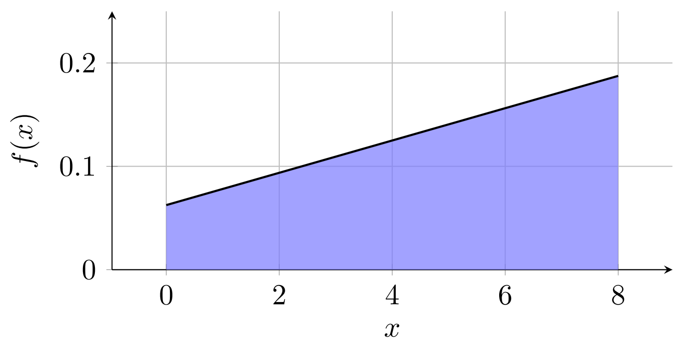
$$\frac{\frac{1}{16}+\frac{3}{16}}{2}\cdot c = 1.$$

So

$$\frac{4/16}{2}\cdot c = 1$$

$$\frac{1}{8}c = 1 \implies c = 8.$$

C1 interpretation



The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. Using $c = 8$, the density is

$$f(b) = \frac{1}{16} + \frac{b}{64} \quad \text{for } 0 \le b \le 8.$$

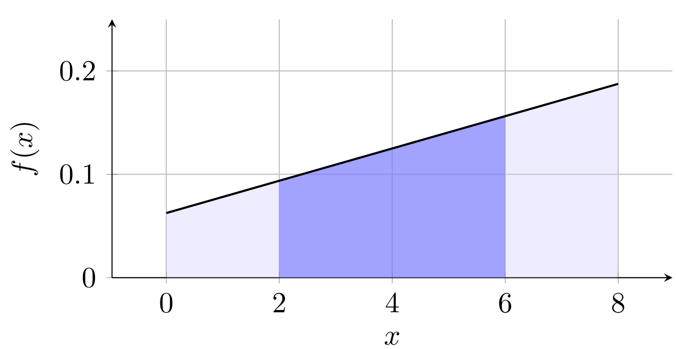
For the comparison:

$$P(0 \le B \le 2) = \frac{\frac{1}{16}+\frac{3}{32}}{2}\cdot 2 = \frac{5}{32}, \qquad P(6 \le B \le 8) = \frac{\frac{5}{32}+\frac{3}{16}}{2}\cdot 2 = \frac{11}{32},$$

so $[6, 8]$ is more likely.

For the requested probability,

$$P(2 \le B \le 6) = \frac{f(2)+f(6)}{2}\cdot 4 = \frac{\frac{3}{32}+\frac{5}{32}}{2}\cdot 4 = \frac{1}{2}.$$



Final answer: $c = 8$ and $P(2 \le B \le 6) = \frac{1}{2}$.

13 Delivery Score (Linear, Both Endpoints Nonzero)

Problem. Let D be a delivery score with support $[-3, c]$ and density

$$f(d) = \frac{1}{40} + \frac{d+3}{5(c+3)} \quad \text{for } -3 \le d \le c.$$

Questions/Tasks.

- **C1 Calculations:** Find c so $f(d)$ is a valid PDF.
- **C1 Interpretation:** Explain why the model is nonnegative on its support.
- **Extra analysis:** Find $P(-1 \le D \le 3)$.

C1

C1 calculations The endpoint values are fixed:

$$f(-3) = \frac{1}{40}, \quad f(c) = \frac{1}{40} + \frac{c+3}{5(c+3)} = \frac{1}{40} + \frac{1}{5} = \frac{9}{40}.$$

Since the graph is a line joining two positive endpoint values, the density is nonnegative on its support.

The total area must be 1. Using the trapezium formula,

$$\frac{\frac{1}{40} + \frac{9}{40}}{2} \cdot (c+3) = 1.$$

So

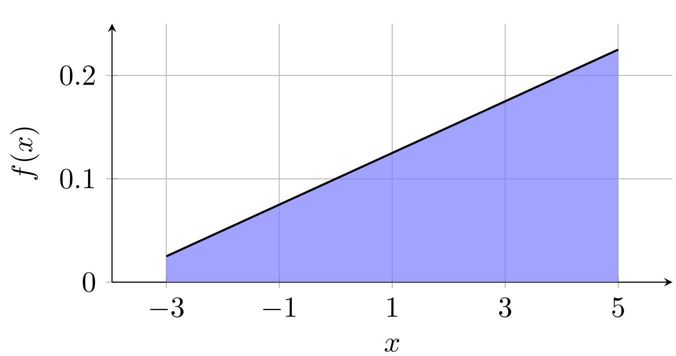
$$\frac{10/40}{2} \cdot (c+3) = 1$$

$$\frac{1}{8}(c+3) = 1 \implies c+3 = 8 \implies c = 5.$$

Now substitute this value into the density:

$$f(d) = \frac{1}{40} + \frac{d+3}{40} = \frac{d+4}{40} \quad \text{for } -3 \le d \le 5.$$

C1 interpretation



The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. For the requested probability,

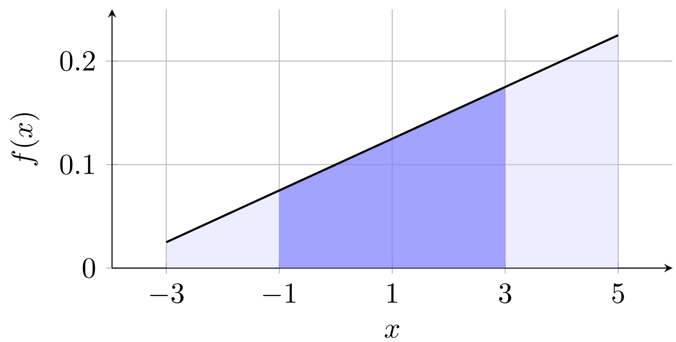
$$P(-1 \le D \le 3) = \frac{f(-1) + f(3)}{2} \cdot 4.$$

With

$$f(-1) = \frac{3}{40}, \quad f(3) = \frac{7}{40},$$

we get

$$P(-1 \le D \le 3) = \frac{\frac{3}{40} + \frac{7}{40}}{2} \cdot 4 = \frac{1}{2}.$$



Final answer: $c = 5$, the density is nonnegative since both endpoint heights are positive and the graph is linear on the support, and $P(-1 \le D \le 3) = \frac{1}{2}$.

14 Clinic Waiting Time (Uniform)

Problem. Let W be clinic waiting time (in min) with support $[2, c]$. The density is a line on this support, with endpoint values

$$f(2) = \frac{1}{8} \quad \text{and} \quad f(c) = \frac{1}{8}.$$

Questions/Tasks.

- **C1 Calculations:** Find c so the model is a valid PDF.
- **C1 Interpretation:** Explain how to compare probabilities of two equal-width waiting-time intervals.
- **Extra analysis:** Find $P(W > 7)$.

C1

C1 calculations Since the density is a line and has the same value at both endpoints, it is constant on the support. So the graph is a rectangle of height $\frac{1}{8}$ and width $c - 2$.

For a valid PDF, the total area must be 1:

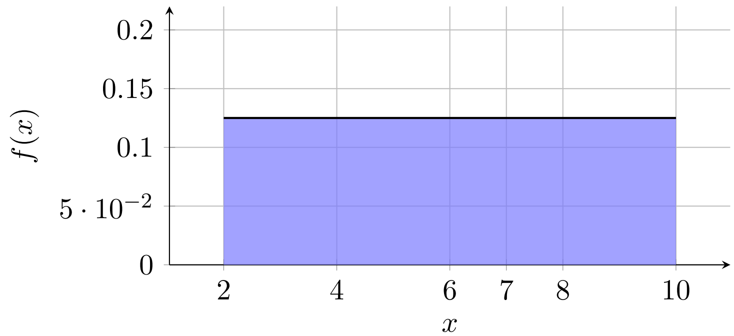
$$(c - 2) \cdot \frac{1}{8} = 1.$$

Thus

$$c - 2 = 8 \implies c = 10.$$

For a constant density, equal-width intervals have equal probability because probability is width $\times \frac{1}{8}$.

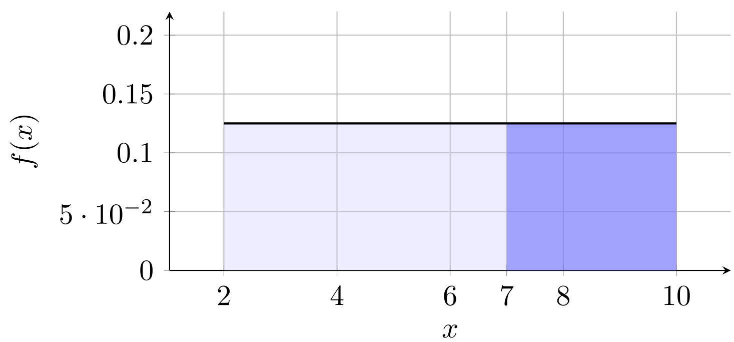
C1 interpretation



The PDF graph shows the support and overall shape of the density on that support.

Extra analysis.

$$P(W > 7) = P(7 < W \leq 10) = (10 - 7) \cdot \frac{1}{8} = \frac{3}{8}.$$



Final answer: $c = 10$ and $P(W > 7) = \frac{3}{8}$.

15 Task Completion Fraction (Linear, Endpoint at 0)

Problem. Let C be a completion fraction with support $[0, b]$. The density is a line on this support, with endpoint values

$$f(0) = 1 \quad \text{and} \quad f(b) = 0.$$

Questions/Tasks.

- **C1 Calculations:** Find b so the model is a valid PDF.
- **C1 Interpretation:** Explain the practical meaning of higher density near $c = 0$ than near the right endpoint.
- **Extra analysis:** Find $P\left(\frac{1}{2} \leq C \leq \frac{3}{2}\right)$.

C1

C1 calculations Since the density is a line with endpoint values $f(0) = 1$ and $f(b) = 0$, the graph is a decreasing triangle with base b and height 1.

For a valid PDF, the total area must be 1:

$$\frac{1}{2}(b)(1) = 1.$$

Thus

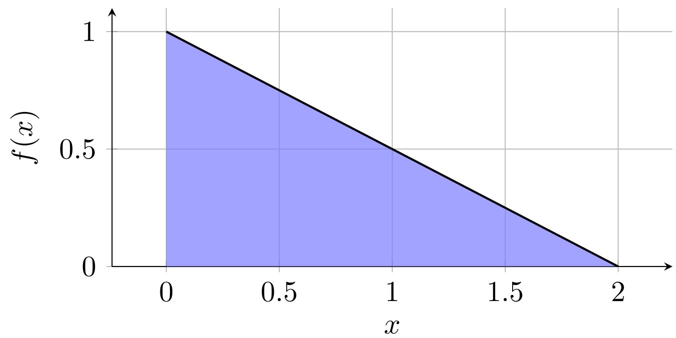
$$b = 2.$$

So the density is the line joining $(0, 1)$ and $(2, 0)$, that is,

$$f(c) = 1 - \frac{c}{2} \quad \text{for } 0 \le c \le 2.$$

Higher density near $c = 0$ means lower completion fractions occur more often than values near full completion.

C1 interpretation



The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. For

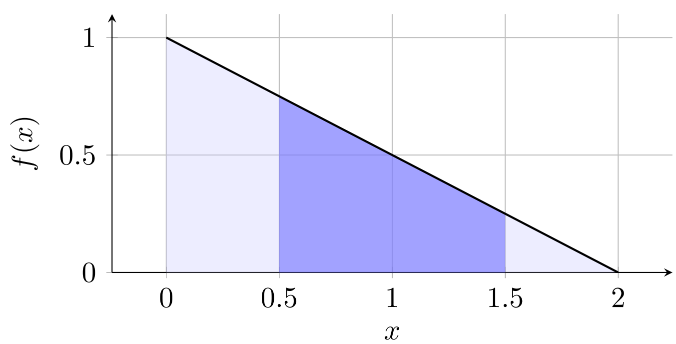
$$\frac{1}{2} \le C \le \frac{3}{2},$$

use trapezium area. The endpoint heights are

$$f\left(\frac{1}{2}\right) = 1 - \frac{1}{4} = \frac{3}{4}, \qquad f\left(\frac{3}{2}\right) = 1 - \frac{3}{4} = \frac{1}{4},$$

with width 1. So

$$P\left(\frac{1}{2} \le C \le \frac{3}{2}\right) = \frac{\frac{3}{4} + \frac{1}{4}}{2} \cdot 1 = \frac{1}{2}.$$



Final answer: $b = 2$ and $P\left(\frac{1}{2} \le C \le \frac{3}{2}\right) = \frac{1}{2}$.

16 Final Performance Measure (Linear, Both Endpoints Nonzero)

Problem. Let P be a performance measure with support $[1, c]$. The density is a line on this support, with endpoint values

$$f(1) = \frac{9}{40} \quad \text{and} \quad f(c) = \frac{1}{40}.$$

Questions/Tasks.

- **C1 Calculations:** Find c so the model is a valid PDF.
- **C1 Interpretation:** Compare $P(1 \le P \le 3)$ and $P(7 \le P \le 9)$ without integration, using geometric reasoning.
- **Extra analysis:** Find $P(3 \le P \le 8)$.

C1

C1 calculations Since the density is a line with endpoint values

$$f(1) = \frac{9}{40}, \quad f(c) = \frac{1}{40},$$

the graph is a decreasing trapezium on $[1, c]$.

For a valid PDF, the total area must be 1:

$$\frac{\frac{9}{40} + \frac{1}{40}}{2} \cdot (c - 1) = 1.$$

Thus

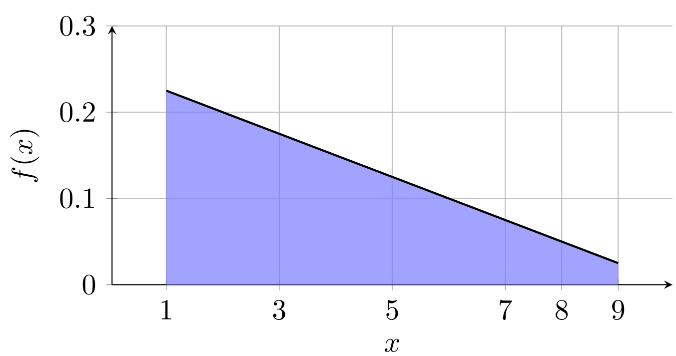
$$\frac{10/40}{2} \cdot (c - 1) = 1$$

$$\frac{1}{8}(c - 1) = 1 \implies c - 1 = 8 \implies c = 9.$$

So the density is the line joining $(1, \frac{9}{40})$ and $(9, \frac{1}{40})$, that is,

$$f(p) = \frac{10 - p}{40} \quad \text{for } 1 \le p \le 9.$$

C1 interpretation



The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. Compare intervals geometrically:

$$P(1 \le P \le 3) = \frac{f(1) + f(3)}{2} \cdot 2$$

with

$$f(1) = \frac{9}{40}, \quad f(3) = \frac{7}{40}.$$

So

$$P(1 \le P \le 3) = \frac{\frac{9}{40} + \frac{7}{40}}{2} \cdot 2 = \frac{16}{40} = \frac{2}{5}.$$

Also,

$$P(7 \le P \le 9) = \frac{f(7) + f(9)}{2} \cdot 2$$

with

$$f(7) = \frac{3}{40}, \quad f(9) = \frac{1}{40}.$$

So

$$P(7 \le P \le 9) = \frac{\frac{3}{40} + \frac{1}{40}}{2} \cdot 2 = \frac{4}{40} = \frac{1}{10}.$$

Therefore,

$$P(1 \le P \le 3) > P(7 \le P \le 9).$$

For $3 \le P \le 8$:

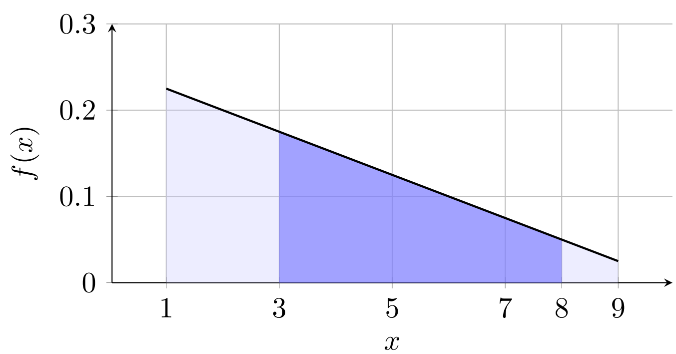
$$P(3 \le P \le 8) = \frac{f(3) + f(8)}{2} \cdot 5$$

with

$$f(3) = \frac{7}{40}, \quad f(8) = \frac{2}{40}.$$

Thus

$$P(3 \le P \le 8) = \frac{\frac{7}{40} + \frac{2}{40}}{2} \cdot 5 = \frac{9}{16}.$$



Final answer: $c = 9$, $P(1 \le P \le 3) = \frac{2}{5} > P(7 \le P \le 9) = \frac{1}{10}$, and $P(3 \le P \le 8) = \frac{9}{16}$.

17 Sensor Error Level (Uniform with Variable Domain)

Problem. Let S be a sensor error level with support $[0, a]$, where $a > 0$, and density

$$f(s) = k \quad \text{for } 0 \le s \le a.$$

Questions/Tasks.

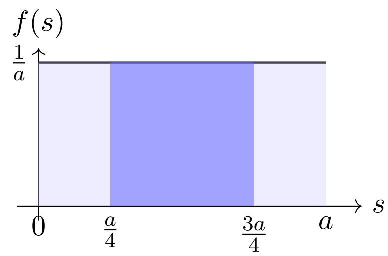
- **C1 Calculations:** Find k in terms of a so $f(s)$ is a valid PDF.
- **C1 Interpretation:** Explain what this model says about equal-length intervals inside $[0, a]$.
- **Extra analysis:** Find $P\left(\frac{a}{4} \leq S \leq \frac{3a}{4}\right)$.

C1

C1 calculations Uniform on $[0, a]$ with $a > 0$ gives

$$ak = 1 \implies k = \frac{1}{a}.$$

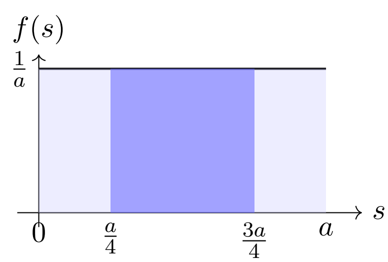
C1 interpretation



The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. For $\frac{a}{4} \leq S \leq \frac{3a}{4}$:

$$P\left(\frac{a}{4} \leq S \leq \frac{3a}{4}\right) = \left(\frac{3a}{4} - \frac{a}{4}\right)\frac{1}{a} = \frac{a/2}{a} = \frac{1}{2}.$$



Final answer: $k = \frac{1}{a}$ and $P\left(\frac{a}{4} \leq S \leq \frac{3a}{4}\right) = \frac{1}{2}$.

18 Traffic Flow Index (Linear with Variable Domain)

Problem. Let F be a traffic flow index with support $[0, b]$, where $b > 0$, and density

$$f(f) = kf \quad \text{for } 0 \leq f \leq b.$$

Questions/Tasks.

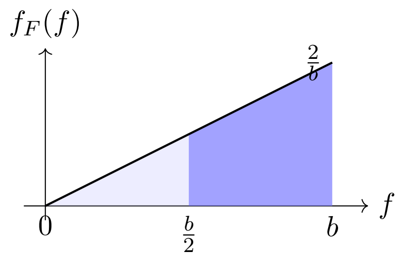
- **C1 Calculations:** Find k in terms of b so $f(f)$ is a valid PDF.
- **C1 Interpretation:** Explain why larger values of f are more likely than smaller values in equal-width intervals.
- **Extra analysis:** Find $P\left(\frac{b}{2} \leq F \leq b\right)$.

C1

C1 calculations For $f(f) = kf$ on $[0, b]$ $(b > 0)$, area is a triangle with base b and height kb:

$$\frac{1}{2}(b)(kb) = \frac{kb^2}{2} = 1 \implies k = \frac{2}{b^2}.$$

C1 interpretation



The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. For the requested probability,

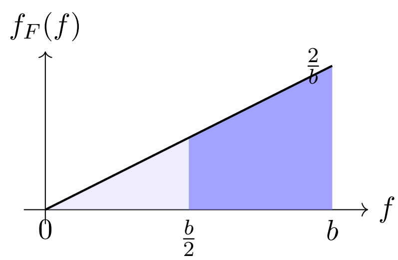
$$P\left(\frac{b}{2} \le F \le b\right) = 1 - P\left(0 \le F \le \frac{b}{2}\right).$$

Left part is a triangle with base $\frac{b}{2}$ and height $k\frac{b}{2}$:

$$P\left(0 \le F \le \frac{b}{2}\right) = \frac{1}{2}\left(\frac{b}{2}\right)\left(k\frac{b}{2}\right) = \frac{kb^2}{8} = \frac{1}{4}.$$

Hence

$$P\left(\frac{b}{2} \le F \le b\right) = 1 - \frac{1}{4} = \frac{3}{4}.$$



Final answer: $k = \frac{2}{b^2}$ and $P\left(\frac{b}{2} \le F \le b\right) = \frac{3}{4}$.

19 Machine Output Deviation (Linear with One Variable Endpoint)

Problem. Let Z be a machine output deviation with support $[2, c]$, where $c > 2$, and density

$$f(z) = k(z - 2) \quad \text{for } 2 \le z \le c.$$

Questions/Tasks.

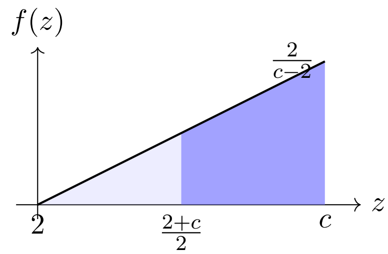
- **C1 Calculations:** Find k in terms of c so $f(z)$ is a valid PDF.
- **C1 Interpretation:** Explain why values near c are more likely than values near 2.
- **Extra analysis:** Find $P\left(\frac{2+c}{2} \le Z \le c\right)$ using geometric reasoning.

C1

C1 calculations For $f(z) = k(z-2)$ on $[2, c]$ $(c > 2)$, area is a triangle with base $(c-2)$ and height $k(c-2)$:

$$\frac{1}{2}(c-2)\,k(c-2) = \frac{k(c-2)^2}{2} = 1 \implies k = \frac{2}{(c-2)^2}.$$

C1 interpretation



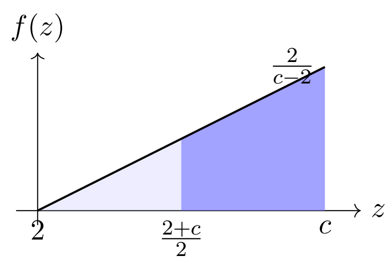
The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. Let midpoint be $\frac{2+c}{2}$. On this interval,

$$f\left(\frac{2+c}{2}\right) = k\frac{c-2}{2}, \quad f(c) = k(c-2), \quad \text{width} = \frac{c-2}{2}.$$

So

$$P\left(\frac{2+c}{2} \le Z \le c\right) = \frac{k\frac{c-2}{2} + k(c-2)}{2} \cdot \frac{c-2}{2} = \frac{3k(c-2)^2}{8} = \frac{3}{4}.$$



Final answer: $k = \frac{2}{(c-2)^2}$ and $P\left(\frac{2+c}{2} \le Z \le c\right) = \frac{3}{4}$.

20 Inspection Time (Uniform with Two Variable Endpoints)

Problem. Let I be an inspection time with support $[a, b]$, where $a < b$, and density

$$f(i) = k \quad \text{for } a \le i \le b.$$

Questions/Tasks.

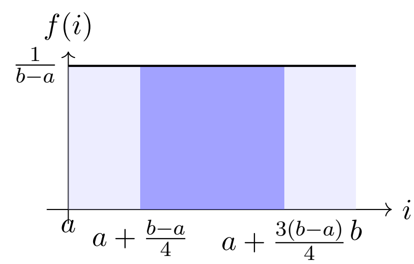
- **C1 Calculations:** Find k in terms of a and b so $f(i)$ is a valid PDF.
- **C1 Interpretation:** Explain why any two equal-length intervals inside $[a, b]$ have the same probability.
- **Extra analysis:** Find $P\left(a + \frac{b-a}{4} \le I \le a + \frac{3(b-a)}{4}\right)$.

C1

C1 calculations Uniform on $[a, b]$ with $a < b$ gives support length $(b - a)$, so

$$k(b-a) = 1 \implies k = \frac{1}{b-a}.$$

C1 interpretation



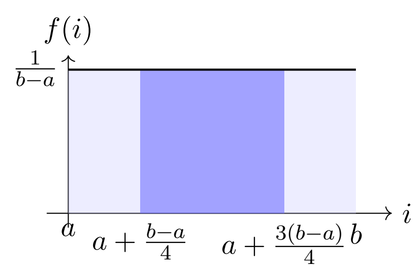
The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. For $a + \frac{b-a}{4} \le I \le a + \frac{3(b-a)}{4}$, interval length is

$$\left(a + \frac{3(b-a)}{4}\right) - \left(a + \frac{b-a}{4}\right) = \frac{b-a}{2}.$$

Hence

$$P = \frac{b-a}{2} \cdot \frac{1}{b-a} = \frac{1}{2}.$$



Final answer: $k = \frac{1}{b-a}$ and $P\left(a + \frac{b-a}{4} \le I \le a + \frac{3(b-a)}{4}\right) = \frac{1}{2}$.

21 Load Measure (Increasing Linear with Two Variable Endpoints)

Problem. Let L be a load measure with support $[m, n]$, where $m < n$, and density

$$f(l) = k(l - m) \quad \text{for } m \le l \le n.$$

Questions/Tasks.

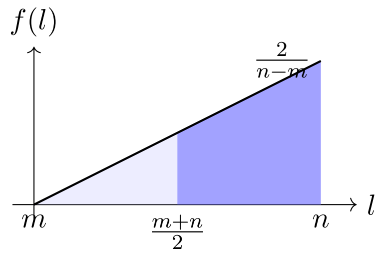
- **C1 Calculations:** Find k in terms of m and n so $f(l)$ is a valid PDF.
- **C1 Interpretation:** Explain why right-side intervals of the same width have more probability than left-side intervals.
- **Extra analysis:** Find $P\left(\frac{m+n}{2} \le L \le n\right)$ using geometric reasoning.

C1

C1 calculations For $f(l) = k(l-m)$ on $[m,n]$ $(m<n)$, this is a triangle from height 0 to height $k(n-m)$.

$$\frac{1}{2}(n-m)\,k(n-m) = \frac{k(n-m)^2}{2} = 1 \implies k = \frac{2}{(n-m)^2}.$$

C1 interpretation



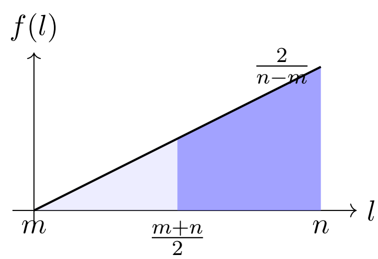
The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. For midpoint interval $\frac{m+n}{2} \le L \le n$:

$$f\left(\frac{m+n}{2}\right) = k\frac{n-m}{2}, \quad f(n) = k(n-m), \quad \text{width} = \frac{n-m}{2}.$$

So

$$P\left(\frac{m+n}{2} \le L \le n\right) = \frac{k\frac{n-m}{2} + k(n-m)}{2} \cdot \frac{n-m}{2} = \frac{3k(n-m)^2}{8} = \frac{3}{4}.$$



Final answer: $k = \frac{2}{(n-m)^2}$ and $P\left(\frac{m+n}{2} \le L \le n\right) = \frac{3}{4}$.

22 Cooling Level (Decreasing Linear with Two Variable Endpoints)

Problem. Let C be a cooling level with support $[r,s]$, where $r<s$, and density

$$f(c) = k(s-c) \quad \text{for } r \le c \le s.$$

Questions/Tasks.

- **C1 Calculations:** Find k in terms of r and s so $f(c)$ is a valid PDF.
- **C1 Interpretation:** Explain why values near r are more likely than values near s.
- **Extra analysis:** Find $P\left(r \le C \le \frac{r+s}{2}\right)$ using geometric reasoning.

C1

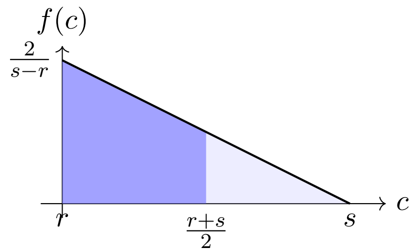
C1 calculations For $f(c) = k(s - c)$ on $[r, s]$ $(r < s)$, heights are

$$f(r) = k(s - r), \quad f(s) = 0,$$

so total area (triangle) is

$$\frac{1}{2}(s - r)\, k(s - r) = \frac{k(s - r)^2}{2} = 1 \implies k = \frac{2}{(s - r)^2}.$$

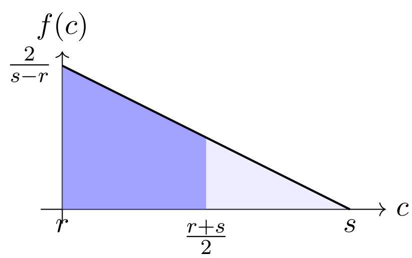
C1 interpretation



The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. For $r \le C \le \frac{r+s}{2}$, width is $\frac{s-r}{2}$ and heights are $k(s - r)$ and $k\frac{s-r}{2}$.

$$P\left(r \le C \le \frac{r + s}{2}\right) = \frac{k(s - r) + k\frac{s-r}{2}}{2} \cdot \frac{s - r}{2} = \frac{3k(s - r)^2}{8} = \frac{3}{4}.$$



Final answer: $k = \frac{2}{(s-r)^2}$ and $P\big(r \le C \le \frac{r+s}{2}\big) = \frac{3}{4}$.

23 Delivery Deviation (Shifted Linear with Two Variable Endpoints)

Problem. Let D be a delivery deviation with support $[p, q]$, where $p < q$, and density

$$f(d) = k(d - p + 1) \quad \text{for } p \le d \le q.$$

Questions/Tasks.

- **C1 Calculations:** Find k in terms of p and q so $f(d)$ is a valid PDF.
- **C1 Interpretation:** Explain why the density stays positive throughout the support.
- **Extra analysis:** Find $P\big(p + \frac{q-p}{3} \le D \le q\big)$.

C1

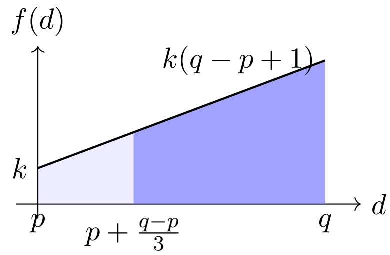
C1 calculations For $f(d) = k(d - p + 1)$ on $[p, q]$ $(p < q)$:

$$d - p + 1 \in [1,\, q - p + 1],$$

so density is positive on the whole support. Let $t = q - p > 0$. Then heights are k and $k(t + 1)$, width t, so

$$\frac{k + k(t+1)}{2} \cdot t = \frac{k(t+2)t}{2} = 1 \implies k = \frac{2}{(q-p)(q-p+2)}.$$

C1 interpretation



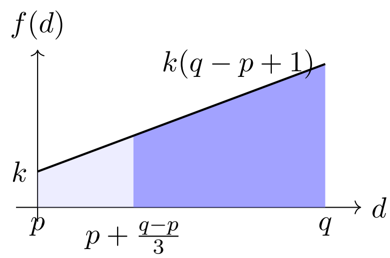
The PDF graph shows the support and overall shape of the density on that support.

Extra analysis. Now compute $P\big(p + \frac{q-p}{3} \le D \le q\big)$. Using $y = d - p$ $(0 \le y \le t)$, this is $y \in \big[\frac{t}{3}, t\big]$ with density $k(y + 1)$:

$$P = \int_{t/3}^{t} k(y+1)\,dy = \frac{k}{2}\left[(y+1)^2\right]_{t/3}^{t} = \frac{k}{2}\left((t+1)^2 - \left(\frac{t}{3}+1\right)^2\right).$$

Substitute $k = \frac{2}{t(t+2)}$ and simplify:

$$P = \frac{4(2t+3)}{9(t+2)} = \frac{4(2(q-p)+3)}{9(q-p+2)}.$$



Final answer: $k = \frac{2}{(q-p)(q-p+2)}$ and

$$P\left(p + \frac{q-p}{3} \le D \le q\right) = \frac{4(2(q-p)+3)}{9(q-p+2)}.$$